

Arnav Nethula

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Education

Rutgers University

Bachelor of Computer Science and Data Science | Minor Mathematics

Expected Graduation: May 2027

New Brunswick, NJ

- **GPA: 3.98** | Member of Dean's List
- **Courses and Organizations:** Deep Learning, Algorithms, Computer Architecture, Data Management, Databases, Discrete Math, Computational Robotics, Linear Algebra, Multivariable Calculus | Rutgers Economics Labs, Rutgers AI, USACS
- **Certifications:** [AWS Certified Cloud Practitioner](#), [AWS Certified AI Practitioner](#), AWS Educate Machine Learning

Work Experience

AT&T

Software Engineering Intern

June 2026 – August 2026

Middletown, NJ

- Developed an automated ticketing flow using **Jira REST API** and creating custom **MCP servers**, enhancing the story creation process to under **10 seconds** with dynamic routing using **Jira fields** and **automation rules**, reducing manual effort by **85%**
- Led development on an **Agentic AI** platform leveraging **LLMs**, **MCP servers**, and **cloud APIs** to analyze infrastructure inventories and generate migration recommendations, reducing manual assessment effort by **53%** across **500+** applications.
- Designed **DDL/ETL** pipelines over **10M+** cost records for **Azure**, **AWS Outposts**, **GCP**, **OCI**, **MongoDB** to **Snowflake**
- Developed **Outposts** cost pipeline with **Snowflake ETL** and vCPU pricing models, exposing **\$72K** in **EC2** and **EBS** spend
- Engineered scalable **anomaly detection** in **Databricks** leveraging telemetry metrics across cloud providers, integrated automated monitoring and remediation workflows through **CI/CD** pipelines and cloud-native **APIs**, reducing operational overhead by **30%**

NeuralSeek

AI Intern

August 2025 – Sep 2025

Remote

- Led development on **5 AI Agents** and **NLP** pipelines using the **NeuralSeek** multi-agent orchestration platform, improving **query processing** throughput by **35%** and supporting **1000+** enterprise users
- Engineered **serverless** architectures leveraging **15+ AWS Lambda** functions, **API Gateway**, **Sagemaker**
- Integrated **LLMs** with enterprise **knowledge databases** through **RESTful APIs** for analyzing **1000+** user queries
- Designed **CI/CD Pipelines** workflows with **GitHub Actions** and **unit testing**, cutting deployment times by **30%**
- Developed scalable **RAG** and **semantic search** pipelines leveraging **vector embeddings** and enterprise **knowledge bases**, with **LLM** orchestration frameworks, improving response accuracy by **40%** and reducing information retrieval latency by **30%**

IDSWG Oncology Group

Software Engineering / AI Developer

May 2025 – Sep 2025

New Brunswick, NJ

- Developed **3** unique **RAG** pipelines for storing patient data utilizing **sentence transformers** and **Ollama LLM**
- Designed a **Streamlit**-based web dashboard on **AWS EC2** for real-time **visualization** and document summarization
- Achieved **68%** data management efficiency and **50%** improvement in summarizing documents optimizing **Mistral-7B**
- Optimized **SQL** queries and schemas on **PostgreSQL**, enabling faster document retrieval and improving database scalability
- Led development of a **Deep Learning** diagnostic by leveraging **Convolutional-Neural-Networks** and **ConvNeXt V2** models to analyze **7000+** radiology scans utilizing **Pandas** and for data cleaning and **PyTorch** for ML achieving **93%+** accuracy

Projects

Agentic AI Economics Ontology | [Github](#)

Python | YAML | Ontology | SQL

- Designed and implemented a **YAML**-driven ontology engine for the Rutgers Economics Labs using **owlread2**, enabling OWL knowledge graphs from heterogeneous **data sources** with zero domain-specific **Python**, reducing schema extension time by **80%**
- Built a modular data **hydration pipeline** through **REST API** and **SQLite**-backed quadstore, optimized query performance via in-memory **caching** and **asynchronous pipeline** orchestration, reducing latency to sub **2** seconds

CVM4X (Hackathon Winner) | [Github](#)

Python | OpenCV | PostgreSQL | Gemini API

- Engineered multi-camera system utilizing **YoloV8** with **OpenCV** for real-time object detection, designed **Flask**-based **RESTful API** backend, improved throughput by **40%** through optimized buffering and asynchronous **I/O**
- Utilized **Gemini API** for **LLM** activity summarization with **PostgreSQL** database and **ChromaDB** vector search for semantic search retrieval, implemented **database indexing** and **data pipelines** to achieve alerts under **200ms** latency

Technical Skills

Languages: Python, Java, Typescript, HTML/CSS, Swift, R, SQL, C

Frameworks / Databases: Spring Boot, Flask, React.JS, Snowflake, MongoDB, PostgreSQL, Streamlit

Libraries: Pandas, Numpy, Matplotlib, Scikit-Learn, Tensorflow, PyTorch, OpenCV2

Developer Tools: AWS, Azure, Databricks, GCP, Oracle Cloud, Git, GitHub, Docker, Jira, Supabase, PowerBI